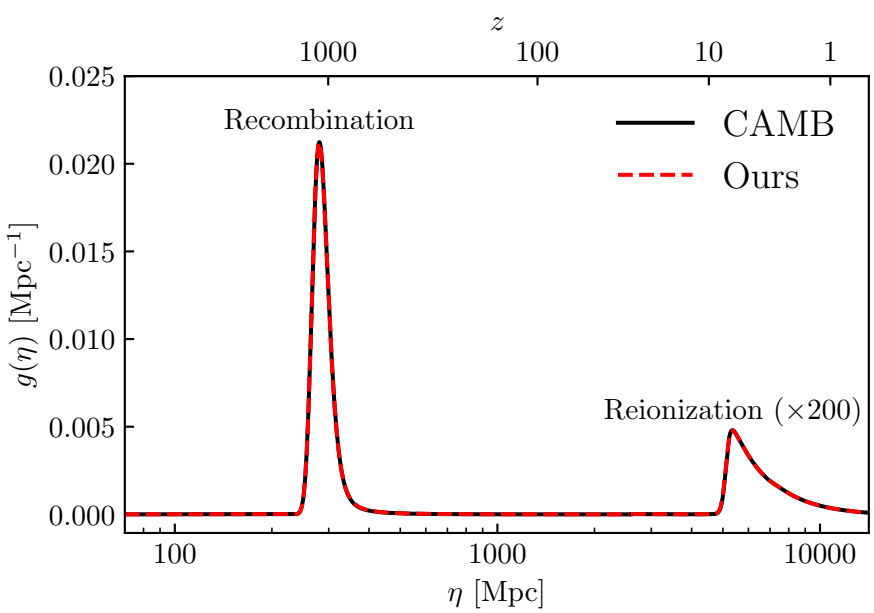
Boltzmann Code - Reionization

Last week in `recombination.pdf` we found that the residual ionization fraction left over from recombination is tiny, $x_e \sim 10^{-4}$. However, observations of quasar absorption spectra tell us that the intergalactic medium is actually highly ionized at $z < 6$ (see derivation of Gunn-Peterson optical depth in `lecture_notes/section03.pdf`). The goal of today's lab is to implement a phenomenological model of reionization to include in the ionization history we built last week, and to use it to compute the two observables: the **optical depth** $\tau(\eta)$ and the **visibility function** $g(\eta)$. At the end of this lab you should be able to recreate the following plot:



Assignment: The rate at which a CMB photon Thomson-scatters off free electrons per unit conformal time is the **Thomson opacity**

$$\boxed{\sigma_T\, a(\eta)\, n_e(\eta) \;=\; \sigma_T\, n_H^{(0)}\, \frac{x_e(\eta)}{a^2(\eta)},} \tag{1}$$

where $n_e(\eta) = x_e(\eta)\, n_H^{(0)}(1+z)^3$ is the free-electron number density and $n_H^{(0)} = (1-Y_P)\rho_b/m_p$ is the hydrogen number density today. The **optical depth** accumulated on a photon's path from conformal time η out to today at η_0 is then

$$\boxed{\tau(\eta) \;\equiv\; \int_\eta^{\eta_0} \sigma_T\, a(\eta')\, n_e(\eta')\, d\eta',} \tag{2}$$

and the probability that a photon emitted at η reached us without scattering is $e^{-\tau(\eta)}$. The probability density (in conformal time) for a photon we observe today to have last-scattered at η is the **visibility function**

$$g(\eta) \;\equiv\; \frac{d}{d\eta}\, e^{-\tau(\eta)} \;=\; \sigma_T\, a(\eta)\, n_e(\eta)\, e^{-\tau(\eta)}. \tag{3}$$

Part 1: Write a function `thomson_opacity(z_arr, x_e_arr, bg)` that returns the Thomson rate $\sigma_T\, a\, n_e$ in units of Mpc^{-1} on an input redshift array. Then write a function `optical_depth(eta_arr, tau_dot_arr)` that takes the Thomson opacity already evaluated on a conformal-time array and returns $\tau(\eta)$. (You will need to compute `eta_arr` from `z_arr` with the `conformal_time` function you wrote a while ago and evaluate `thomson_opacity` on that array before calling `optical_depth`.) An easy way to implement `optical_depth` is with `scipy.integrate.cumulative_trapezoid`; just be careful about the direction of integration and the sign (the definition in Eq.(2) runs from η out to η_0, so τ should be a *decreasing* function of η).

Now we need a model for $x_e(z)$ during reionization. The full physics of reionization is very messy and modeling it properly requires a lot of work (see e.g. Gnedin & Madau (2022)). However, for our boltzmann code it is good enough for now to pick a simple smooth parameterization that interpolates between $x_e \approx 0$ at high z and $x_e \approx 1$ after reionization and fit its midpoint $z_{\rm re}$ to the measured $\tau_{\rm reion}$. We'll use the parameterization used by CAMB, introduced in Appendix A of Lewis (2008). Because reionization happens deep in matter domination, we have $H \propto (1+z)^{3/2}$ and so

$$\frac{d\tau}{dz} = \frac{\sigma_T\, n_H^{(0)}\, x_e\, (1+z)^2}{H(z)} \propto x_e \sqrt{1+z},$$

which lets us write Eq.(2) as

$$\tau_{\rm reion} \propto \int x_e\, d\Big[(1+z)^{3/2}\Big]. \tag{4}$$

Eq. (4) motivates parameterizing x_e as a function of $y \equiv (1+z)^{3/2}$, and CAMB's choice is a tanh step in y:

$$\boxed{x_e(z) = \frac{f}{2}\left[1 + \tanh\left(\frac{y(z_{\rm re}) - y(z)}{\Delta y}\right)\right], \quad y(z) \equiv (1+z)^{3/2}.} \tag{5}$$

On top of this hydrogen step, a second tanh of the same functional form is added to model the later double-ionization of helium, HeII $\to$ HeIII, at $z \sim 3.5$. The CAMB defaults, which we will follow, are:

- **y-exponent:** $y(z) \equiv (1+z)^{3/2}$, chosen so that Eq.(4) is linear in y during matter domination.
- **HI + HeI step:** amplitude $f = 1 + f_{\rm He}$, where

$$f_{\rm He} \equiv n_{\rm He}/n_{\rm H} = Y_P/[(m_{\rm He}/m_p)(1-Y_P)] \approx 0.08,$$

so $f \approx 1.08$ for the Planck best-fit Y_P. Width $\Delta z = 0.5$, i.e. $\Delta y = \frac{3}{2}(1+z_{\rm re})^{1/2}\,\Delta z$. (HeI is bundled in with HI because its first ionization energy is very close to that of HI.)

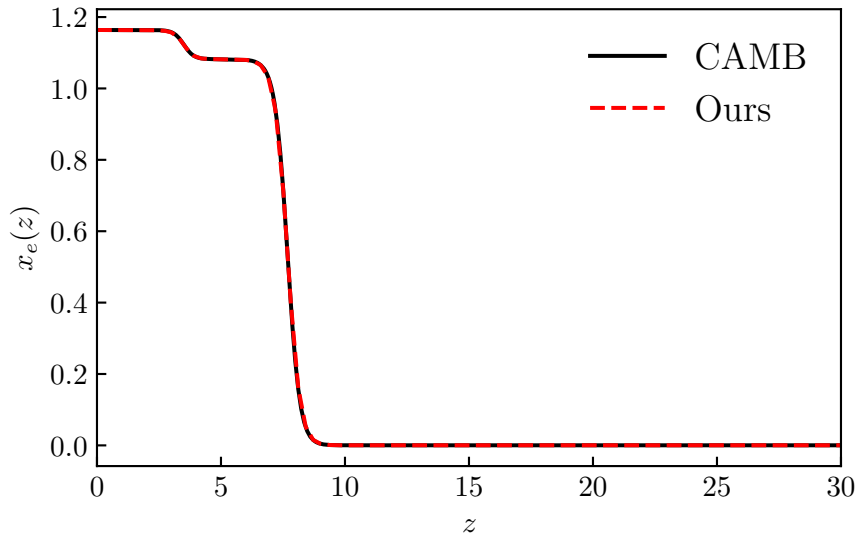
- **HeII step:** amplitude $f_{\rm He}$, centered at $z_{\rm HeII} = 3.5$ with width $\Delta z_{\rm HeII} = 0.4$. This contributes only $\sim 10^{-3}$ to $\tau_{\rm reion}$ but we include it for completeness.

Part 2: Since hydrogen and helium reionization both use the functional form Eq. (5), factor it out into a helper `_tanh_step_in_y(z, z_mid, delta_z, amplitude)`. Then use it to write `reion_xe(z, z_re, f_He)`, which returns the reionization contribution to $x_e(z)$ from the combined HI+HeI step at $z_{\rm re}$ and the HeII step at $z = 3.5$.

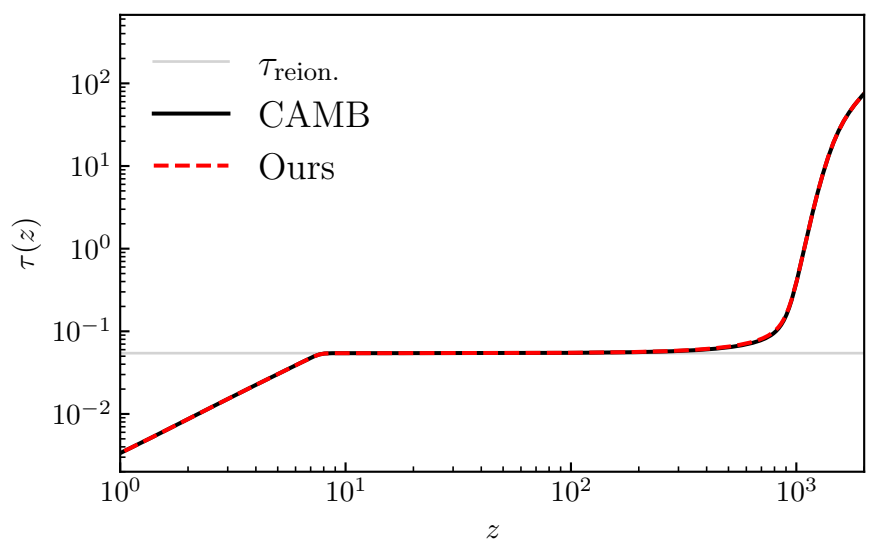
The last ingredient is finding the midpoint redshift $z_{\rm re}$ to match our assumed $\tau_{\rm reion}$. Because $\tau_{\rm reion}^{\rm model}(z_{\rm re})$ is a monotonically increasing function of $z_{\rm re}$, this one-dimensional root-finding problem is cleanly solved by bisection/binary search. If you're unfamiliar, the idea is simple. We start with some loose bound $[z_{\tt lo}, z_{\tt hi}]$ that we expect contains $z_{\rm re}$. We then evaluate $\tau^{\rm model}$ from part 1 given our reionzation model from part 2 assuming the midpoint $z_{\tt mid} = (z_{\tt lo} + z_{\tt hi})/2$ is the reionization redshift. If $\tau^{\rm model}(z_{\tt mid}) > \tau^{\rm target}$ the root must lie in $[z_{\tt lo}, z_{\tt mid}]$; otherwise it lies in $[z_{\tt mid}, z_{\tt hi}]$. After k iterations the bracket has width $(z_{\tt hi} - z_{\tt lo})/2^k$, so bisection converges in $\mathcal{O}(\log_2 n)$ iterations; compare to a linear grid search, which costs $\mathcal{O}(n)$ evaluations to reach the same resolution.

Part 3: Write the function `find_z_re(target_tau, bg)` which binary-searches for the $z_{\rm re}$ such that the optical depth of the reionization model Eq. (5) matches `target_tau`. Then call it with `cosmo['tau_reion']` to obtain $z_{\rm re}$, evaluate `reion_xe` on your full z-grid, and combine it with your Peebles solution from last week using `np.maximum(x_e_reion, xe_arr)`. Taking the maximum (rather than adding) avoids double-counting the small residual from recombination where the reionization tanh floor overlaps with the late-time Peebles tail.

Validation: Let's validate what we've built. The first check is the reionization piece of $x_e(z)$. You can compare with `CAMB`'s `x_e`, with both the combined HI + HeI step at $z_{\rm re}$ and the small HeII double-ionization bump at $z \sim 3.5$ visible.



Next, use your full ionization history that includes both recombination from last week and reionization this week to compute the optical depth $\tau(z)$ back out through recombination. This should track `CAMB`'s `opacity` pretty well:



Finally, putting τ and the Thomson rate together gives the visibility function $g(\eta)$. You should see the narrow recombination peak at $\eta_\star \sim 280$ Mpc (i.e. $z_\star \sim 1090$) and a much smaller and wider reionization bump around $\eta \sim 10^4$ Mpc. The reionization bump is about two orders of magnitude smaller than the recombination peak, so if you want to see both on the same axes you will need to rescale one of them (the plot on the first page of this handout uses a factor of 200).

References:

- Lewis (2008), *Cosmological parameters from WMAP 5-year temperature maps*, 0804.3865.
  Appendix A introduces the tanh-in-$y = (1+z)^{3/2}$ parameterization we use here and explains why it is the natural choice given that $\tau_{\rm reion}$ is a linear functional of $x_e(y)$ in matter domination.

- Gnedin & Madau (2022), *Modeling Cosmic Reionization*, 2208.02260.
  Overview of a bunch of models for reionization like analytical "reionization equation" models, semi-numerical codes like `21cmFAST`, and partially/fully coupled radiation-hydrodynamic simulations.

- Loeb & Furlanetto (2013), *The First Galaxies in the Universe*, Princeton University Press
  Pedagical (though sometimes a bit terse...) treatment of of early galaxies, the IGM, and reionization.